

Basin MCMC Assignment 1

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The purpose of this assignment is to break down the development of the initial basin probability distribution and probability in the Basin MCMC algorithm.

Question 1.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_1$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_1 : \mathcal{X}_1 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. a good approximation to the log posterior in the region of high probability around θ_0 ;
3. easy to compute defining quantities from θ_0 ;
4. easy to sample from;
5. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. **TODO: read over.**

Question 2.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log likelihood $\log p(y | \theta)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $g_2 : \mathcal{X}_2 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $g_2(\theta_1, \theta_2) = g_2(\theta_2, \theta_1)$;
3. a good approximation to the log likelihood in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. **TODO: read over.**

Question 3.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_2 : \mathcal{X}_2 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $f_2(\theta_1, \theta_2) = f_2(\theta_2, \theta_1)$;
3. a good approximation to the log posterior in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant. This should build on the function found in Question 2.

Solution.

Question 4.

[pts.]

Fix $n \geq 2$. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_n$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_n : \mathcal{X}_n \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable;
3. a good approximation to the log posterior in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant, and this constant need not be common to all n . This should be a generalization of the function found in Question 3.

Solution.

Question 5.

[pts.]

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Rewrite the function found in Question 1 in the same form as the functions found in Question 4.

If it is not possible to write the function found in Question 1 in the same form as the functions found in Question 4, explain why. Then, provide an alternative function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_1$ that is in the same form as the functions found in Question 4.

Solution.

Question 6.

[pts.]

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide an approximation of the sequence of probabilities $\{p_n\}_{n \in \mathbb{N}}$ where

$$p_n := P(\theta \in X_n | y) = \int_{\mathcal{X}_n} p(\theta | y) d\theta,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0 .

You may find it useful to use the functions found in Questions 4 and 5. Note that you need only approximate the sequence up to a common multiplicative constant; that is, it suffices to approximate quantities proportional to $\{p_n\}_{n \in \mathbb{N}}$.

Solution.

Question 7.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide an approximation of the probability p where

$$p :=,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0 .

Solution.